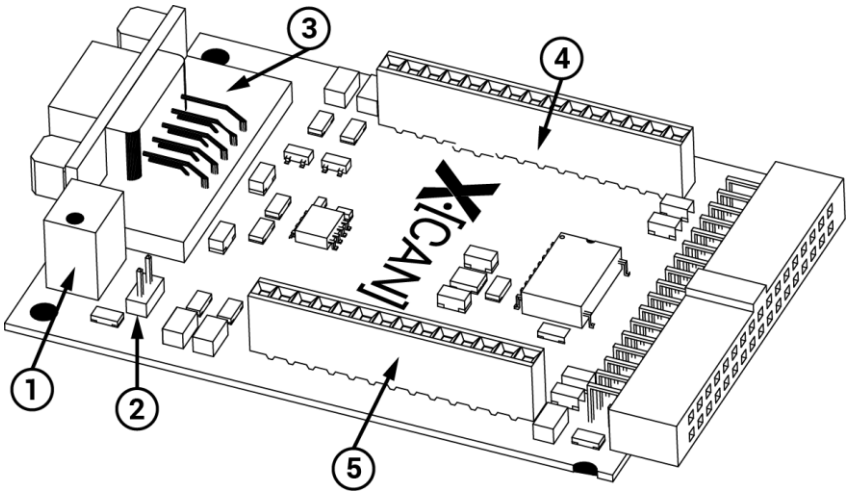
HARDWARE USER GUIDE AND SPECIFICATIONS

X-CAN for myRIO

The X-CAN for myRIO is a CAN 2.0 add-on module for the National Instruments myRIO-1900. This document contains information on device pinouts and specifications for the X-CAN for myRIO.

Figure 1. X-CAN for myRIO



1	3 Pin Screw Terminal	4	I/O Breakout
2	Termination Header	5	I/O Breakout
3	DB-9 Connector		

Safety Information

Caution Before installing your device, read all product documentation to ensure compliance with safety, EMC, and environmental regulations.

Note about CE: X-CAN is sold without an enclosure and therefore “out-of-the-box” does not meet all requirements for CE compliance. However, when tested in a specified enclosure all requirements for CE compliance passed. For more on X-CAN CE testing or enclosure information visit ni.com/myrio/can.

Hardware Overview & Pinouts

The following information and figures describe the hardware and pinouts for the X-CAN for myRIO device.

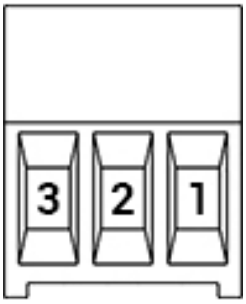
Termination Header (J4)

For high-speed CAN, both ends of the pair of signal wires (CAN_H and CAN_L) must be terminated. This is because communication flows both ways on the CAN bus. CAN_L is pin 2 and CAN_H is pin 7 on the standard 9-pin D-SUB connector. The termination resistors on a cable should match the nominal impedance of the cable. ISO 11898 requires a cable with a nominal impedance of 120 Ohms; therefore, you should use 120 Ohm resistors for termination. If you place multiple devices along the cable, only the devices on the ends of the cable need termination resistors. To enable the 120 Ohm resistor built-into the X-CAN board, place a jumper onto the termination header (as seen in Figure 1). To disable the 120 Ohm resistor, remove the jumper from the termination header.

3 Pin Screw Terminal

The following figure lists the pinouts for the three pin screw terminal block on the X-CAN for myRIO device.

Figure 2. X-CAN for myRIO 3 Pin Screw Terminal

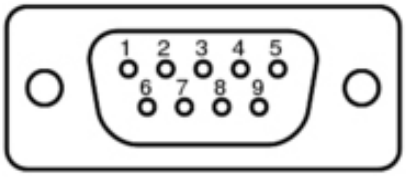


1	CAN High	3	Ground
2	CAN Low		

DB-9 Connector

The following figure lists the pinouts for the DB-9 connector on the X-CAN for myRIO device.

Figure 3. X-CAN for myRIO DB-9 Connector

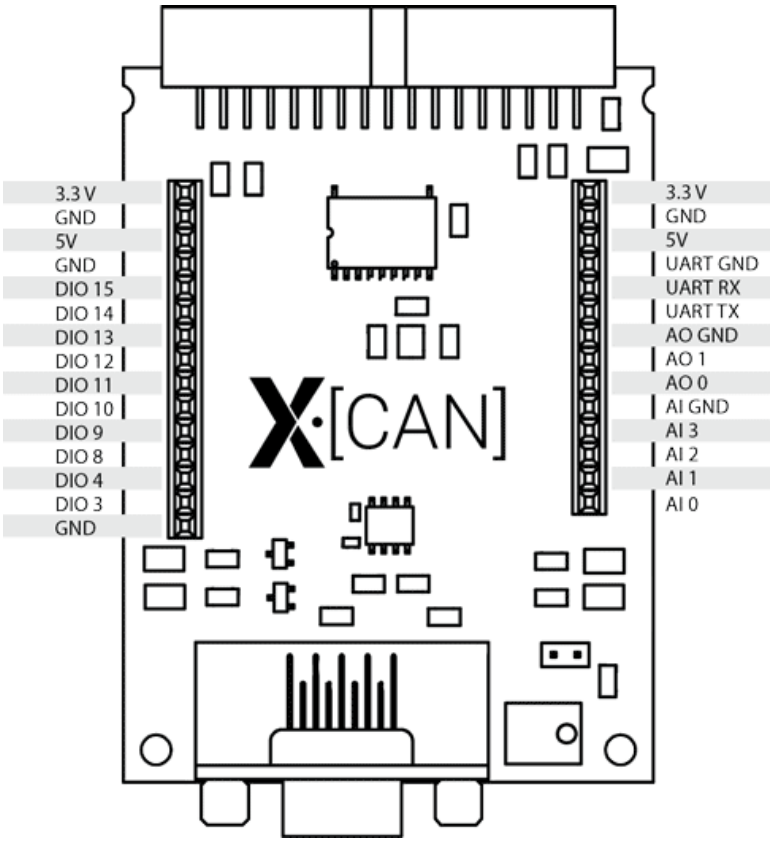


1	Unused	6	Unused
2	CAN Low	7	CAN High
3	Ground	8	Unused
4	Unused	9	Unused
5	Unused		

I/O Breakout

The following figure list the pinouts for the I/O breakout connectors on the X-CAN for myRIO device.

Figure 4: X-CAN for myRIO I/O Breakout



Specifications

The following specifications are typical for the 0 to 40°C operating temperature range unless otherwise noted.

General

User LEDs	Two (Blue)
Mounting	myRIO MXP connector
Interface	High speed SPI (4MHz)
Input Power	+5V @ 100 mA Max (from myRIO)
Output Power	See myRIO MXP power specifications
Temperature Range	0 to 40°C (limited by myRIO)

CAN

Physical Layer	ISO-11898 standard
Common Mode Bus Voltage	-2.0 to +7.0 V
Max DC Voltage on CANH and CANL	-58 to +58 V
Supported Frame Format	CAN2.0 up-to 1Mb/s standard & extended
Receive Buffers	Two w/prioritized message storage
Connectors	CiA DS 102-2 (DB-9) & 3 pin screw terminal

Dimensions & Weight

Dimensions	3.17 x 2.23 x .56 in.
Weight	≈1.2 oz.